FLATNESS OF THE COMPLETION OF A MODULE UNDER FINITE GENERATION AND TOR VANISHING HYPOTHESES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Flatness of the module modulo the ideal propagates to the quotients by all powers. Algebra, Lemma \ref{algebra-lemma-what-does-it-mean}, cited at proof lines 6541-6542. Gives flatness of every quotient by a power of the ideal.
- Completions are adically complete and the quotients by powers of the ideal are unchanged. Algebra, Lemma \ref{algebra-lemma-hathat-finitely-generated}, cited at proof lines 6543-6545. Identify the completed module as the inverse limit of the quotients.
- The completion of a ring with finitely generated ideal and Noetherian quotient is Noetherian. Algebra, Lemma \ref{algebra-lemma-completion-Noetherian}, cited at proof lines 6546-6547. Gives Noetherianity of the completed ring.
- Flatness of an inverse limit of a surjective system of flat modules. Lemma \ref{lemma-limit-flat}, cited at proof lines 6548-6550. Concludes flatness of the completed module.

Source reference. Original lemma 0AGW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13209–13209. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 051C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6530–6530. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05GG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7060–7060. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05GH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7061–7061. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0912: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11430–11430. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let $I \subset R$ be an ideal. Let M be an R-module. Assume*

(1) *I is finitely generated,*
(2) *R/I is Noetherian,*
(3) *M/IM is flat over R/I,*
(4) *$Tor_1^R(M, R/I) = 0$.*

Then the I-adic completion $R^\wedge$ is a Noetherian ring and $M^\wedge$ is flat over $R^\wedge$.

Proof. By Algebra, Lemma 051C the modules M/I^nM are flat over R/I^n for all n. By Algebra, Lemma 05GG we have (a) $R^\wedge$ and $M^\wedge$ are I-adically complete and (b) $R/I^n = R^\wedge/I^nR^\wedge$ for all n. By Algebra, Lemma 05GH the ring $R^\wedge$ is Noetherian. Applying Lemma 0912 we conclude that $M^\wedge = \lim M/I^nM$ is flat as an $R^\wedge$-module. □